

AI in signal and audio processing

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Syllabus

- **Session 1:** Introduction & Fundamentals of Signal/Audio
- **Session 2:** Audio Features & Librosa
- **Session 3:** Audio Processing with Torchaudio
- **Session 4:** Sound Detection and Classification
- **Session 5:** Sequence Modeling with RNNs and LSTMs
- **Session 6:** Automatic Speech Recognition (ASR)
- **Session 7:** Speaker and Emotion Recognition
- **Session 8:** Project presentation

Fundamentals of Signal and Audio Processing

Session 1

Fundamentals of Signal and Audio Processing

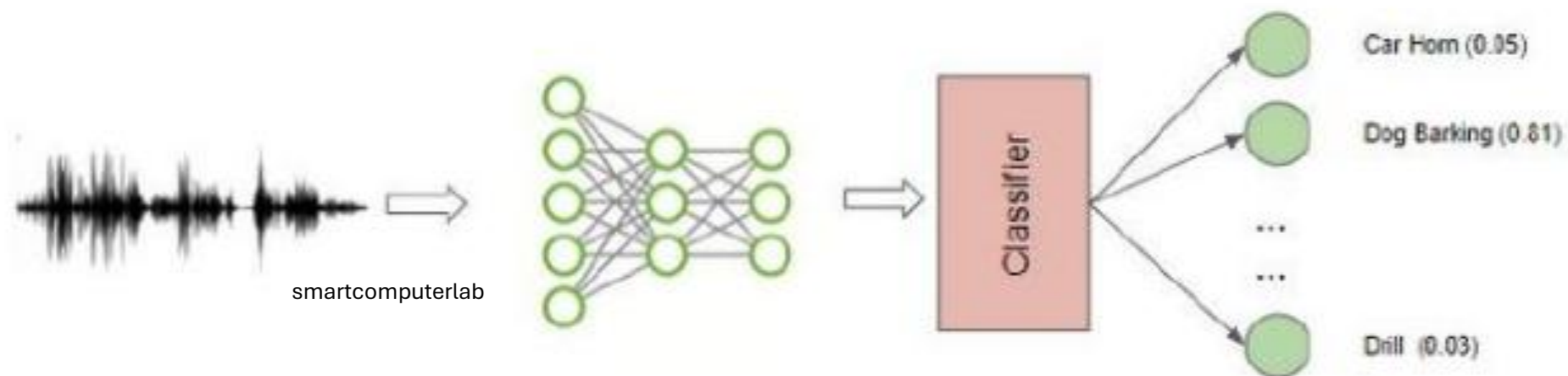
Session 1

Objectives:

- Understand what a signal is and its mathematical properties
- Explain key differences between continuous and discrete signals
- Describe sound as a physical and perceptual phenomenon
- Understand the concept of sampling and digitization
- Visualize time-domain vs. frequency-domain representations

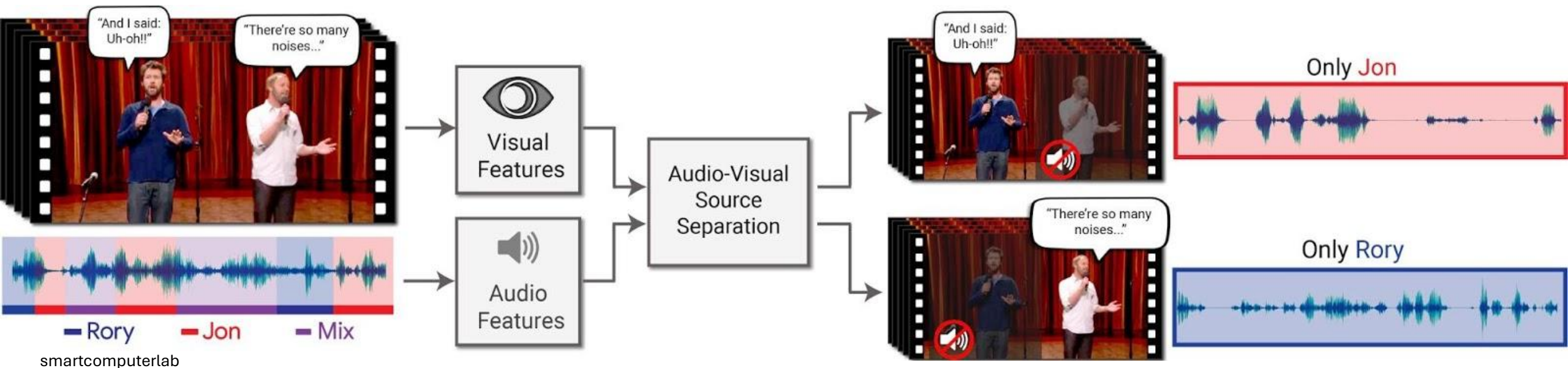
Why audio IA is important/useful for?

- **Classification**



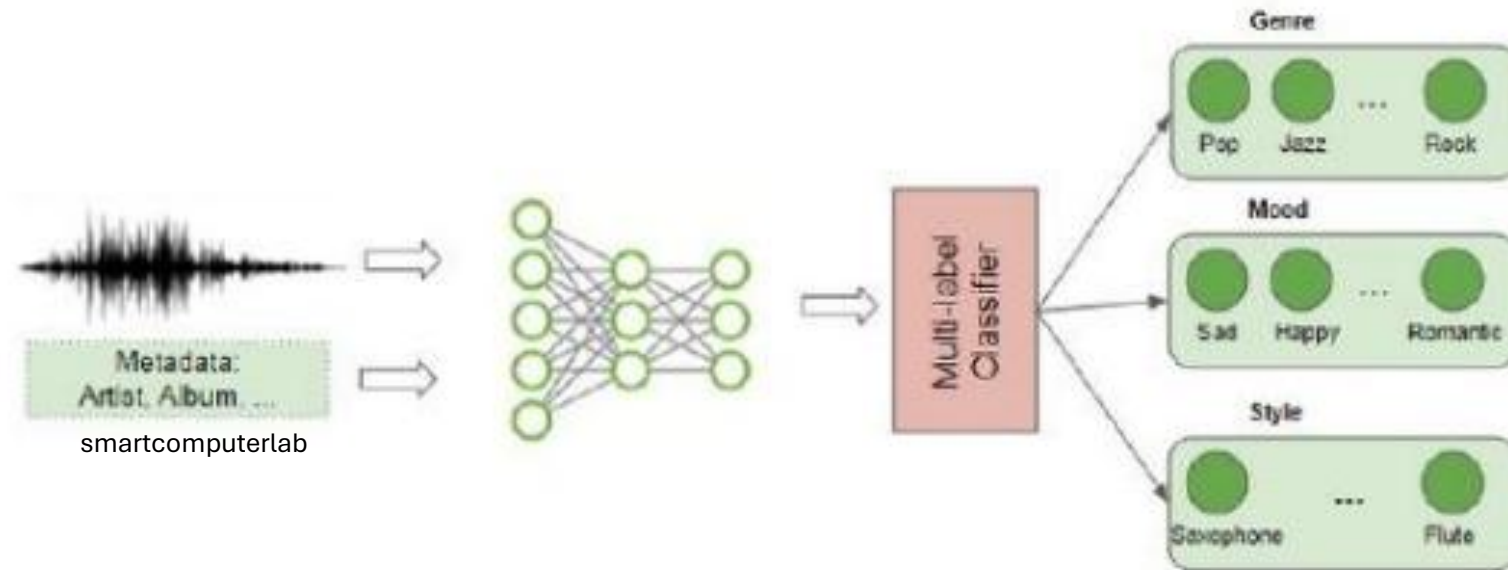
Why audio IA is important/useful for?

- Separation and segmentation



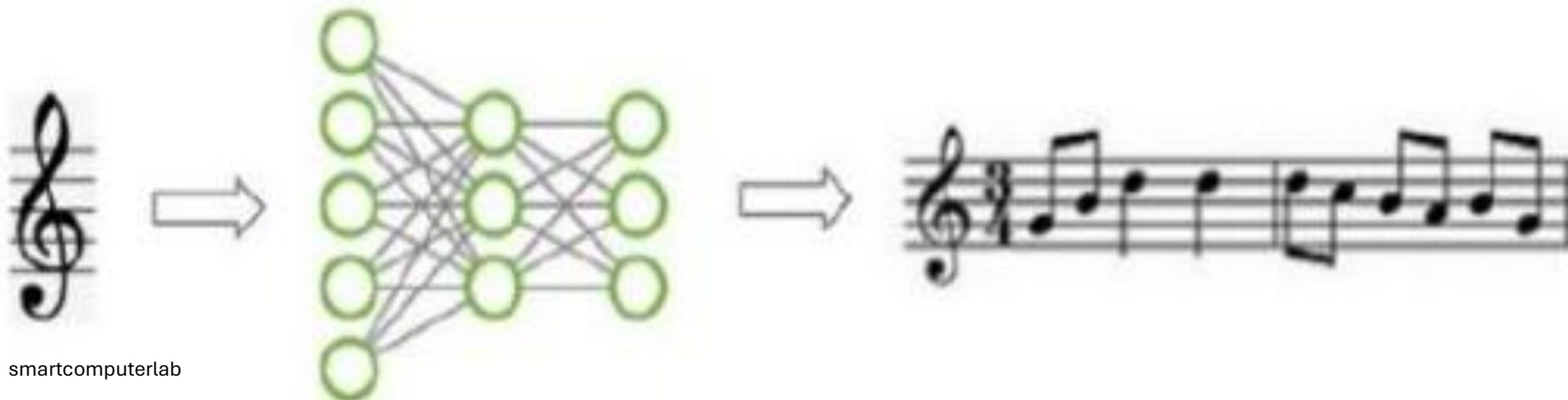
Why audio IA is important/useful for?

- Music genre classification



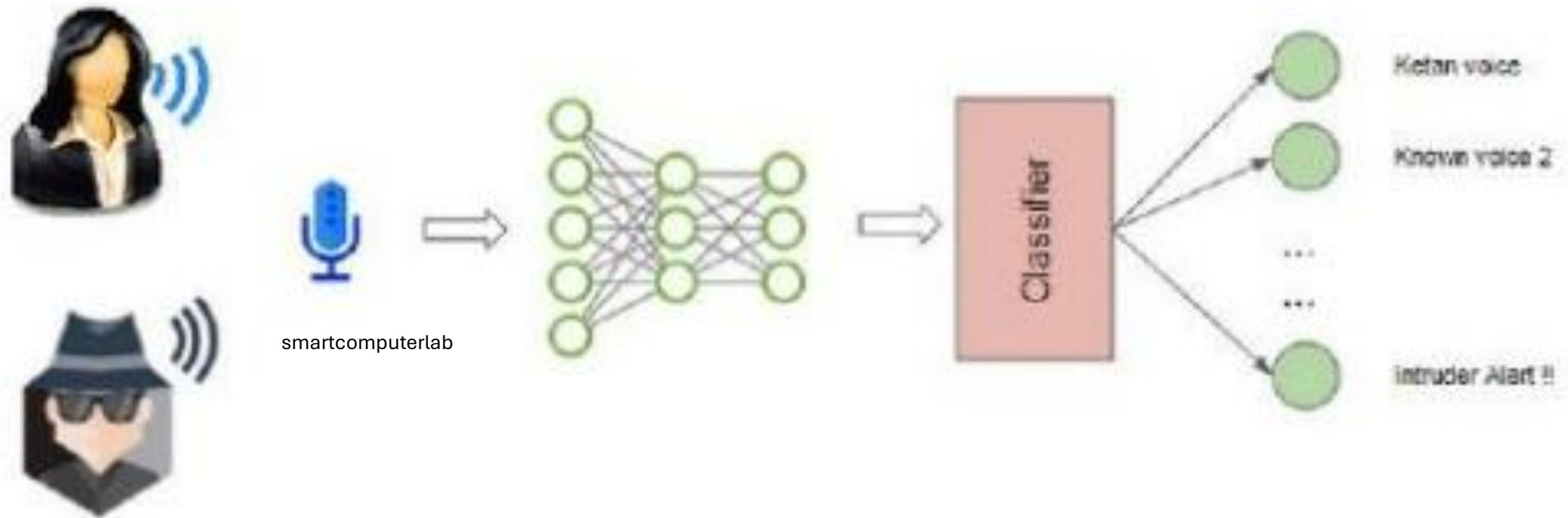
Why audio IA is important/useful for?

- **Music generation**



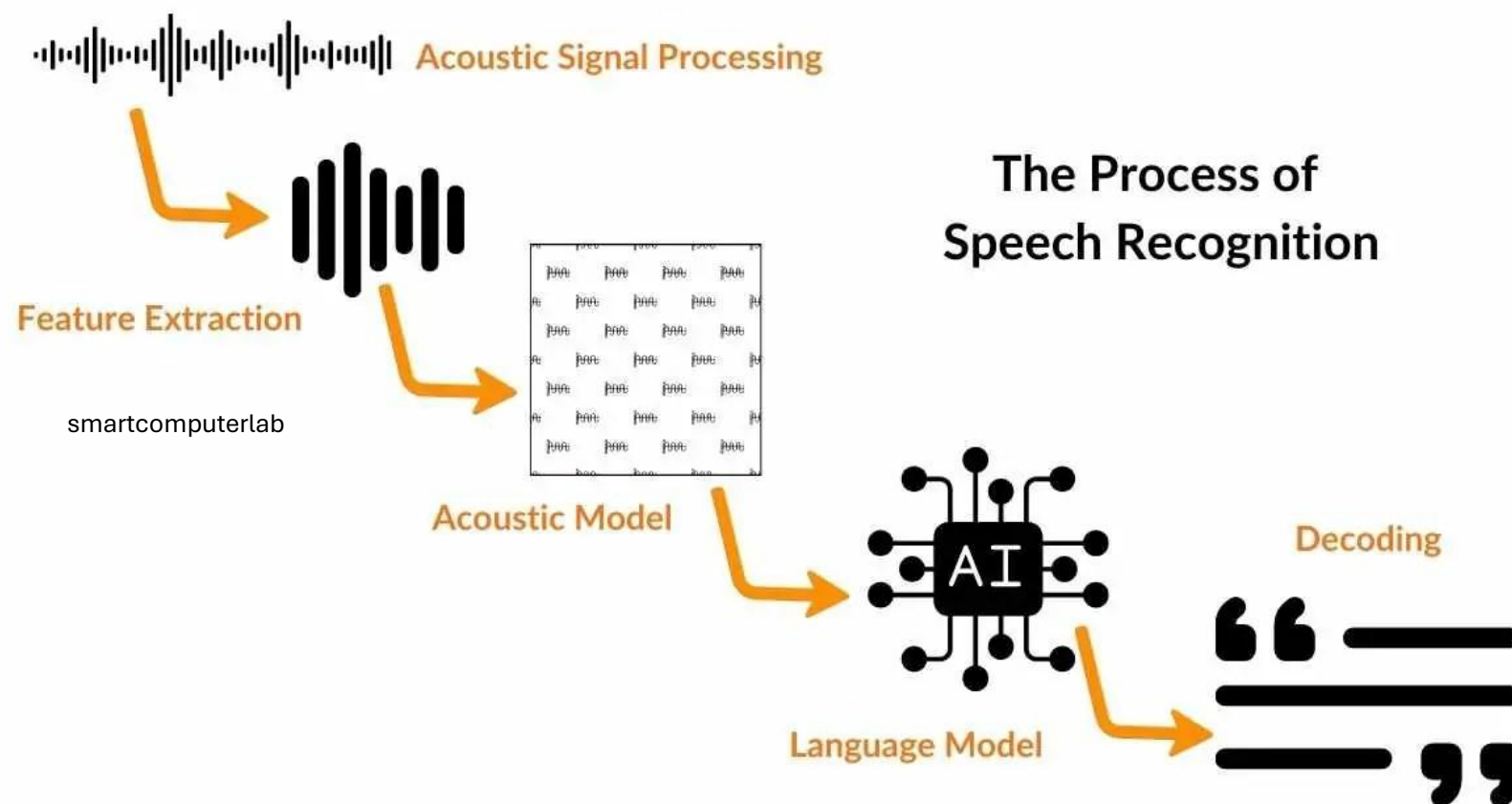
Why audio IA is important/useful for?

- **Voice recognition**



Why audio IA is important/useful for?

- Speech to text



What is Signal and Audio Processing

Definition:

Signal processing is the science of analyzing and manipulating signals—any measurable time-dependent data. Transforming, examining, and interpreting audio signals to extract meaningful information.

Audio signal processing focuses on sound-related signals like speech, music, or noise

Why using IA ?

- Audio data is complex: noisy, variable, context-dependent — ideal for machine learning.
- You can:
 - Analyze thousands of sounds rapidly
 - Detect patterns invisible to humans
 - Eliminate subjectivity in analysis

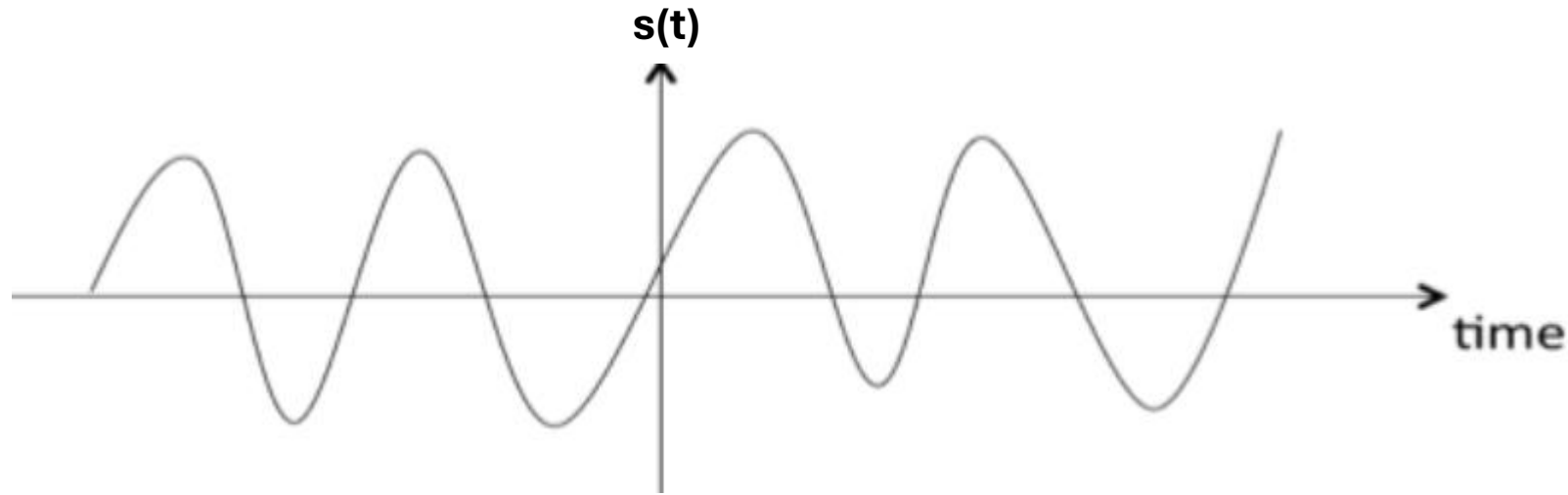
Where is audio processing is used?

Applications across fields:

- Medical (cough detection, heart sounds)
- Music recommendation (Spotify)
- Security (voice recognition)
- Accessibility tools (speech-to-text)
- Telecom, etc

What is a signal?

A signal is a measurable physical quantity which varies as a function of time t . It can be denoted $s(t)$ or $s[n]$

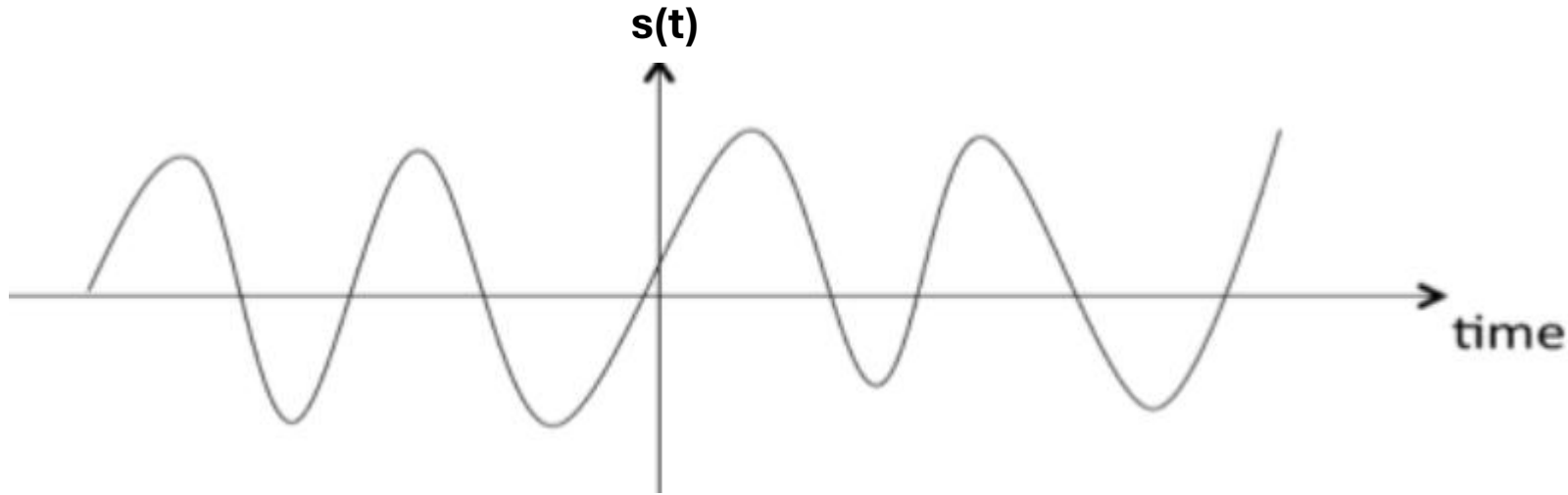


Example:

- Temperature
- Stock price
- Image (2D but a signal)
- ECG ElectroCardioGram

What is a signal?

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Exemple:

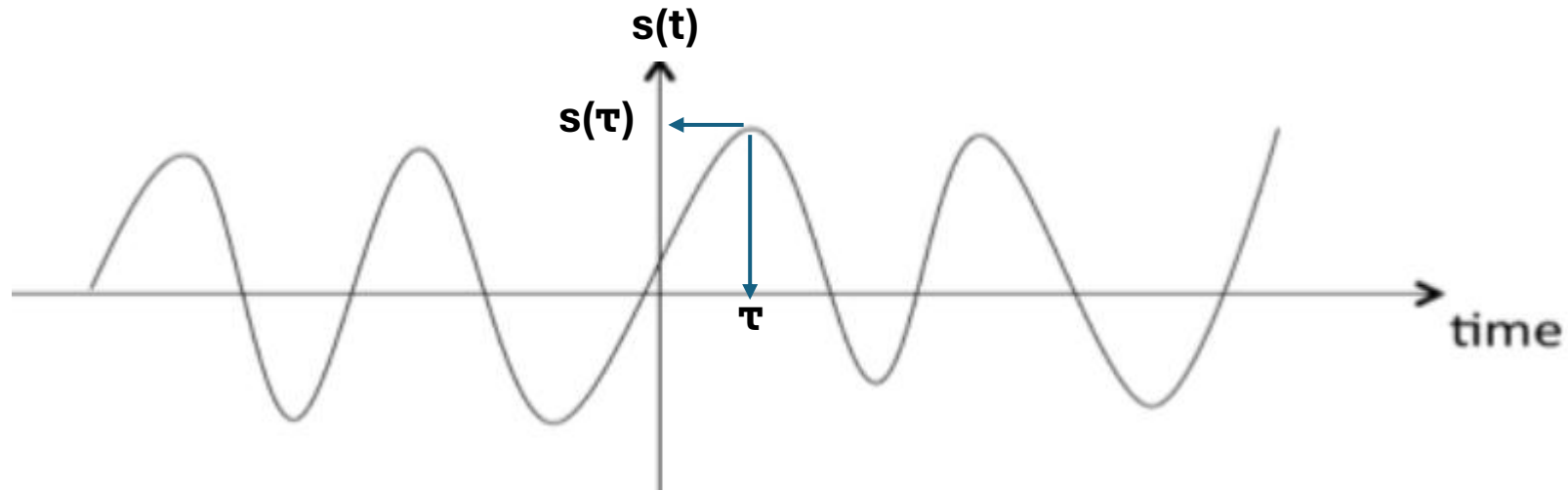
- Temperature
- Stock price
- Image (2D but a signal)
- ECG ElectroCardioGram

Modelling:

- A signal $s(t)$ can be mathematically described by one or more mathematical functions related to each other :
- $\sin(2\pi t)$, $\cos(2\pi t + \pi)$, $\exp(t)$, etc

Amplitude?

The value of the signal at an instant of time τ is called the signal amplitude at time τ .

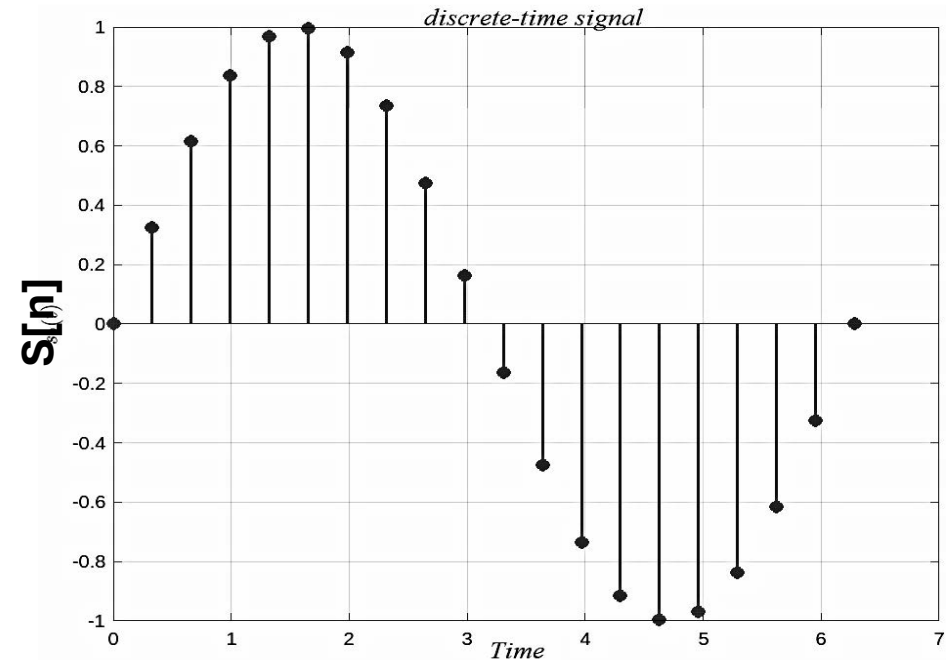
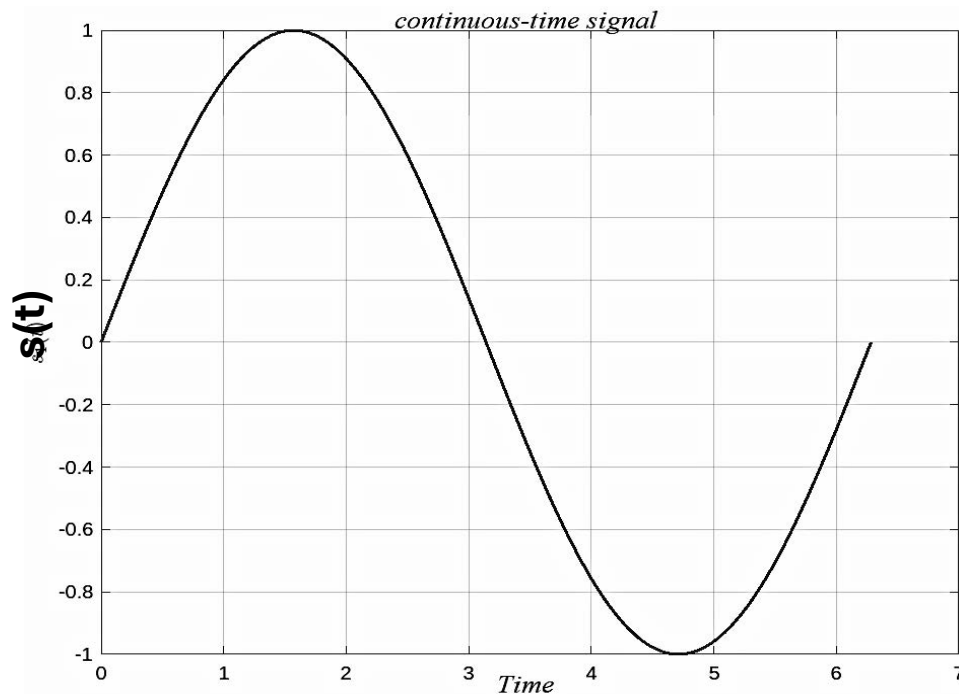


The amplitude of the signal $s(t)$ at time $t = \tau$ is denoted by $s(\tau)$.

Note: If the amplitude does not vary over time, the signal is **constant**

Two type of signals

- **Continuous:** $s(t)$ is a continuous-time signal if it is defined at any time in the interval $[0, t_f]$
- **Discrete:** $s[n]$ is a discrete-time signal if it is defined only at integer time steps n in the interval $[0, t_f]$.

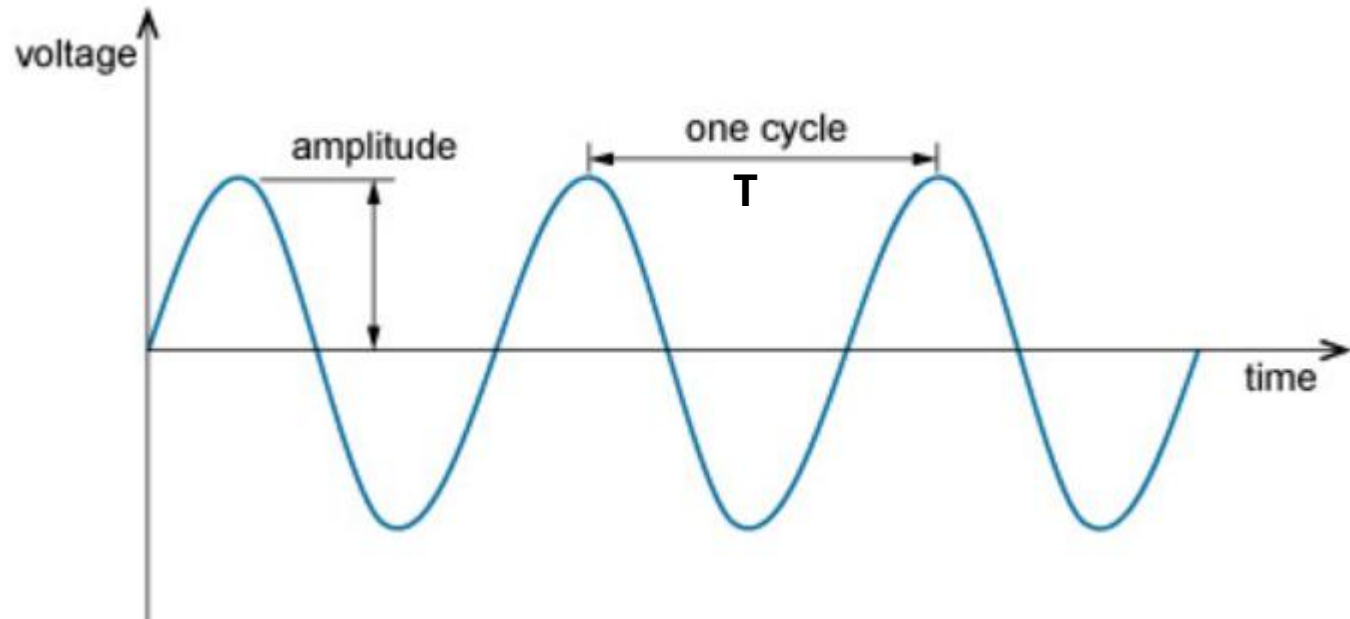


Periodic signal

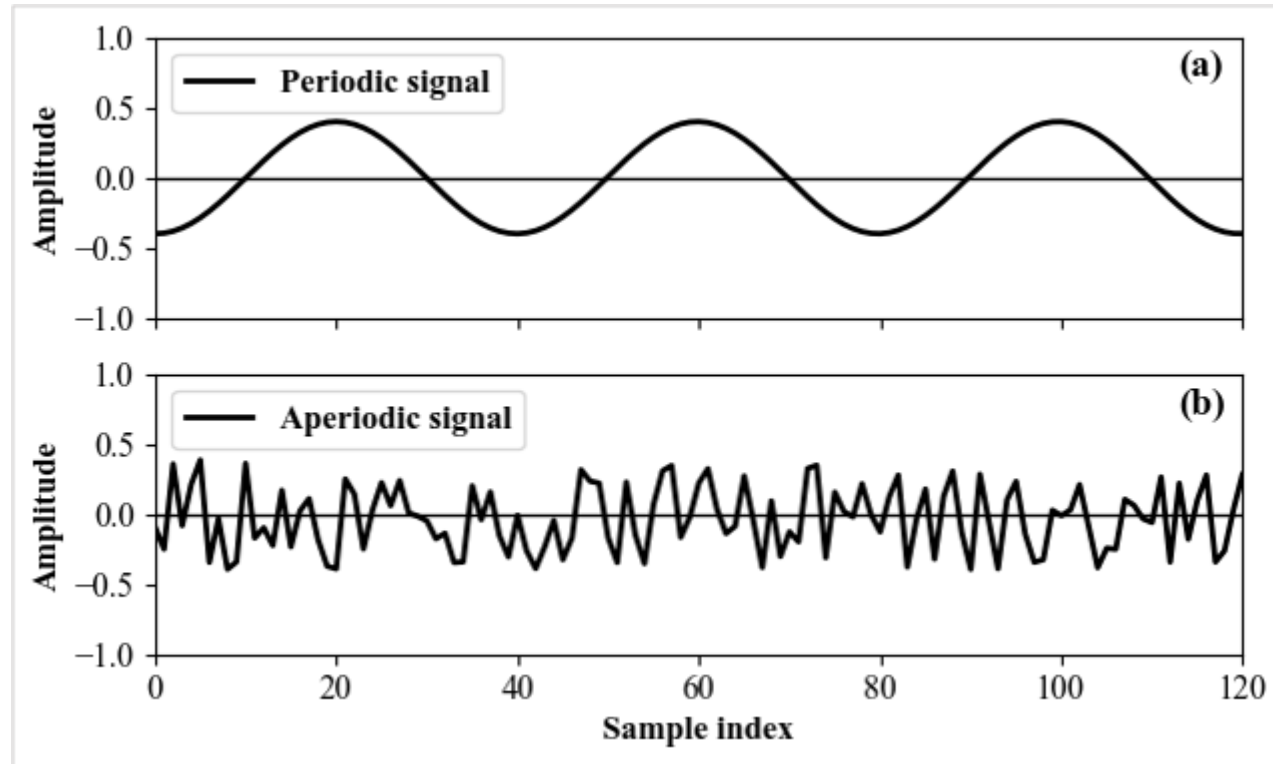
- **Continuous:** if the variations in the amplitude of the signal recur regularly after a constant period T .

$$s(t)=s(t+T)= s(t-T)$$

Frequency: $f=1/T$ (Hz if T in s)



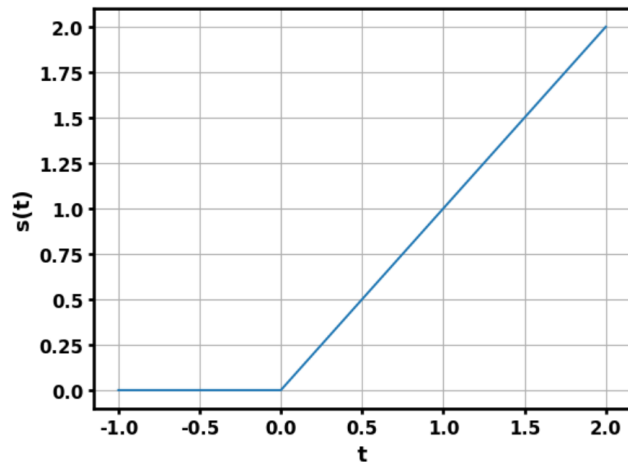
Periodic signal



Signal operation

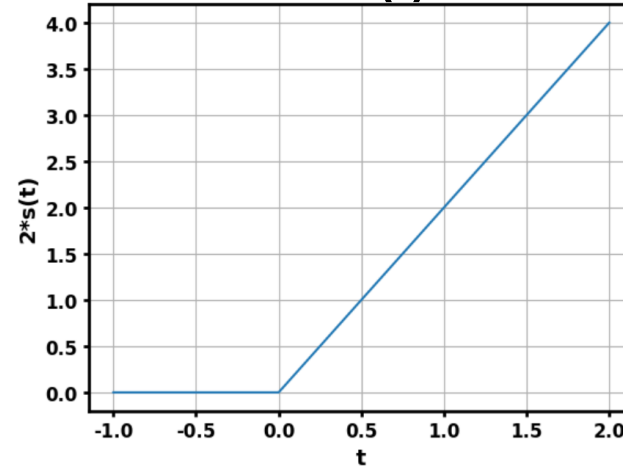
- You can **sum**, **subtract**, **multiply**, or divide two signals:

$s(t)$



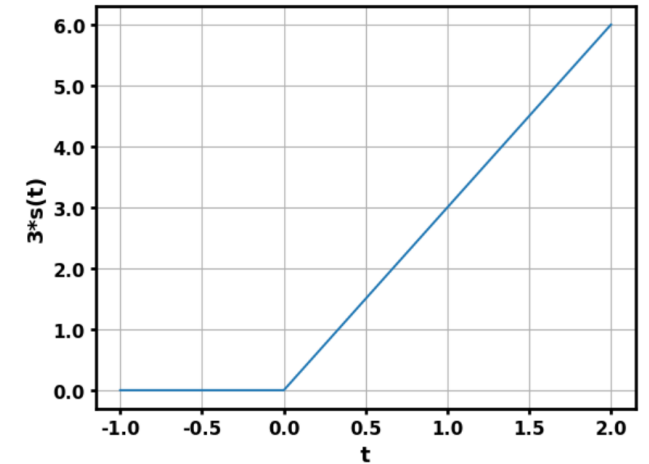
+

$2*s(t)$

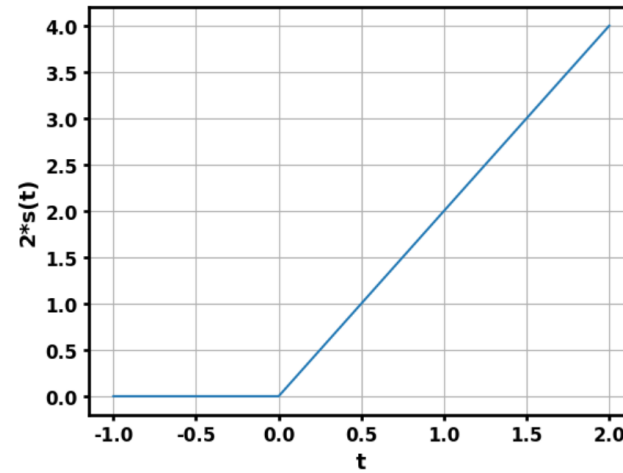


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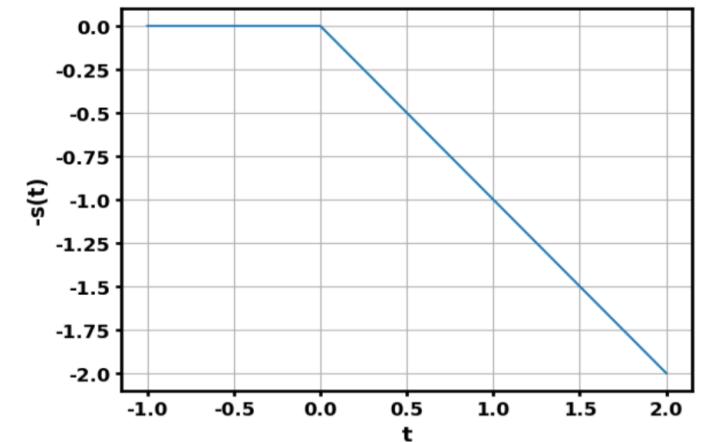
$3*s(t)$



-

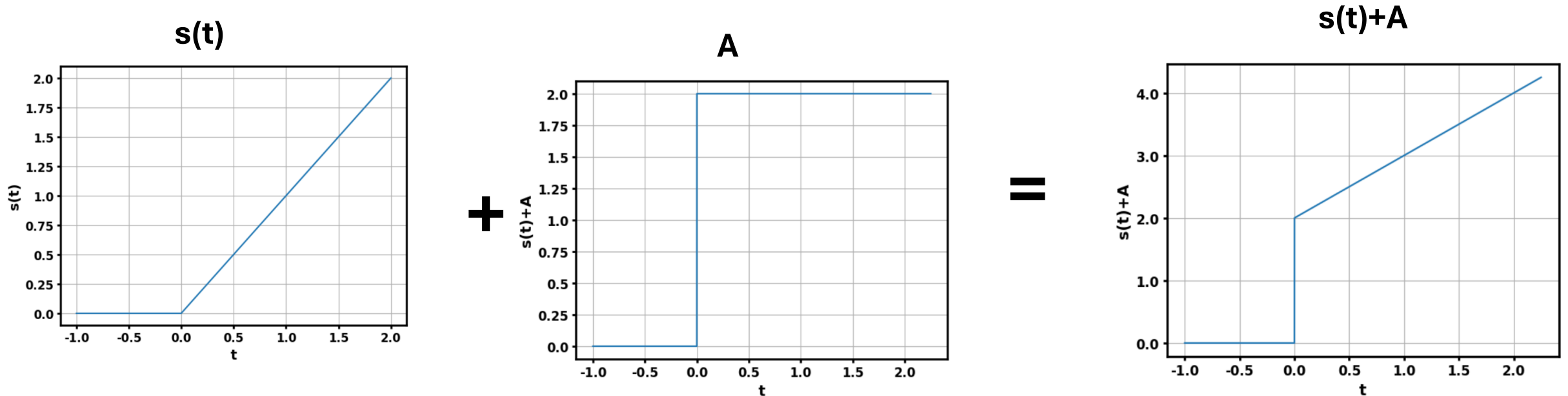


$-s(t)$



Signal operation

- **Offset:** a constant component added to the signal.

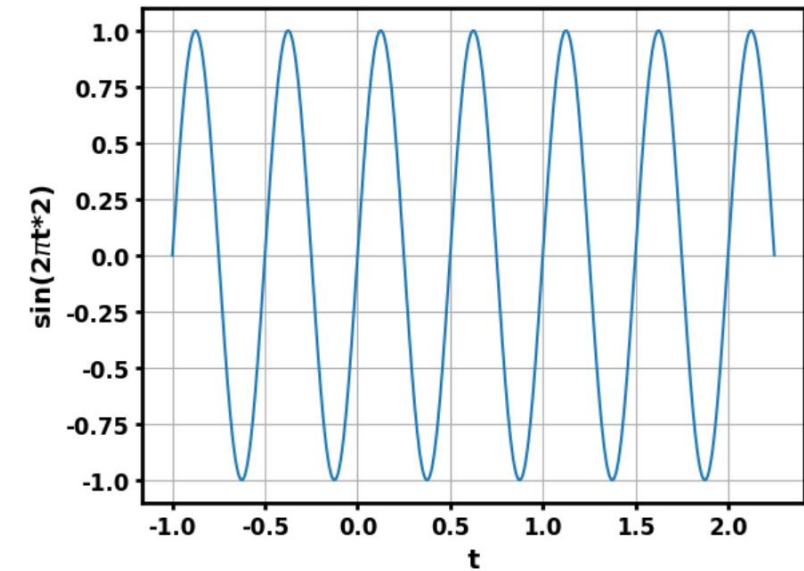
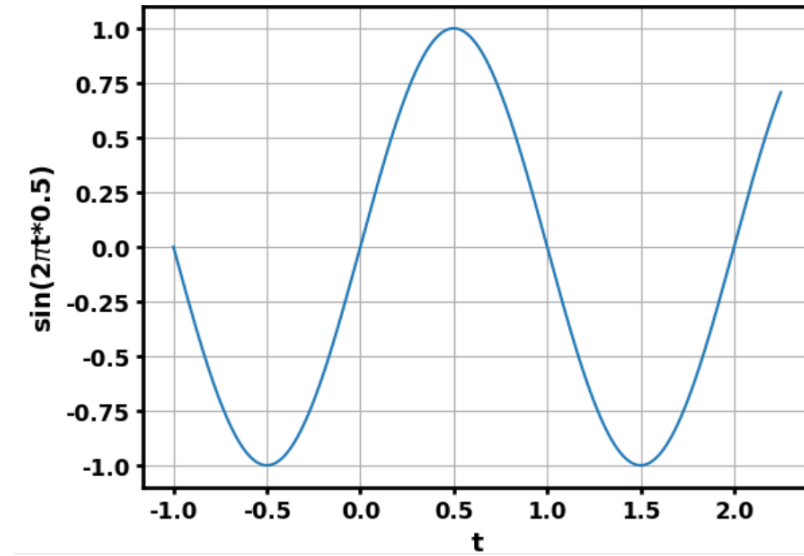
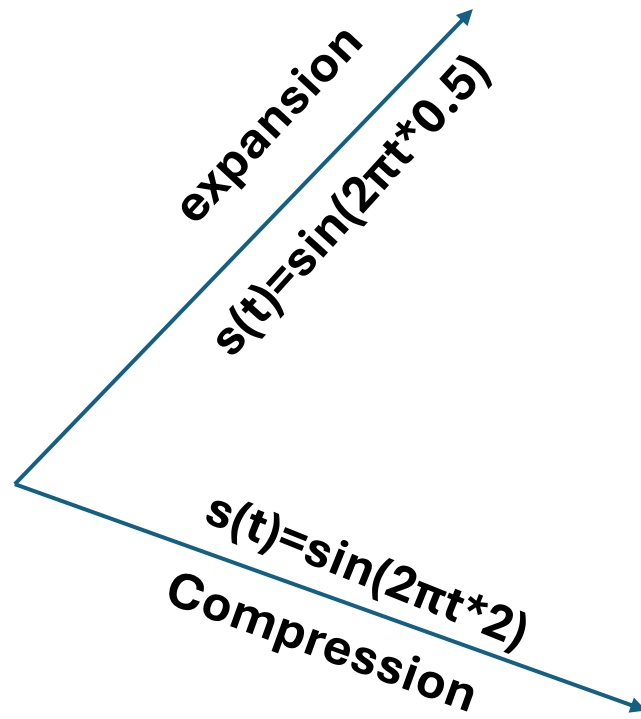
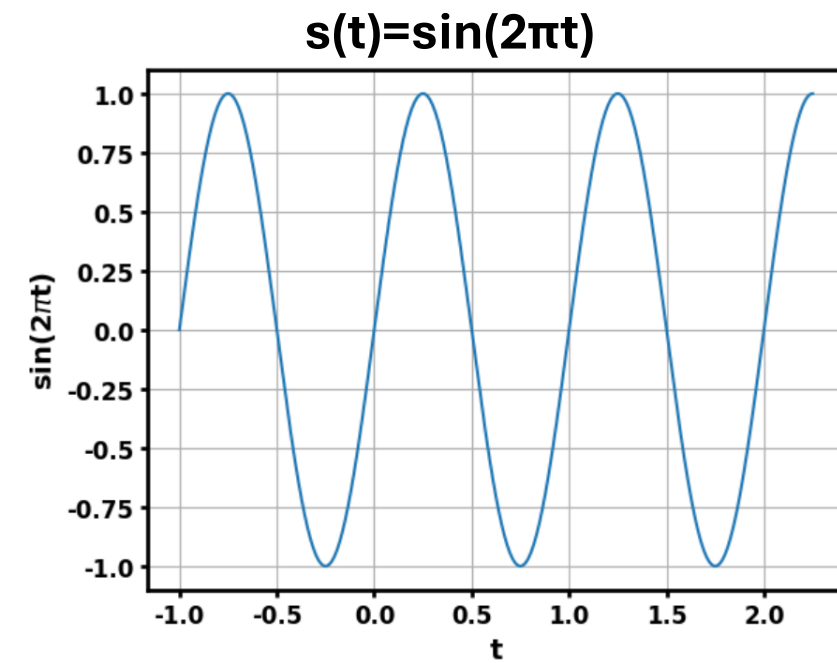


- **Amplification/Attenuation:** Constant to amplify/reduce the amplitude at each instant $s(t)*A$
 - If $A > 1$ = amplification, $0 < A < 1$ = attenuation, if $A < 0$ is a inversion of $s(t)$ with an amplification (if $A < 1$) or an attenuation (if $1 < A < 0$).

Signal operation

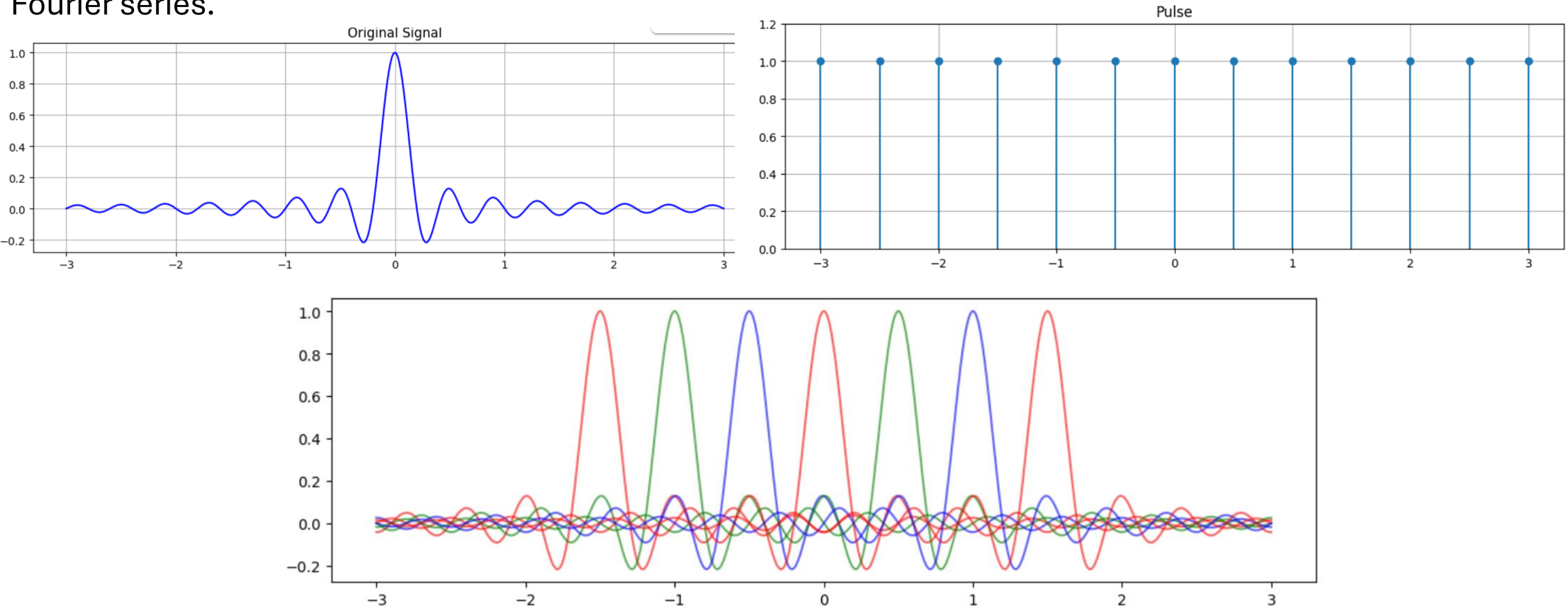
- **Time shift:** use to delay or advance a signal.
 - $s(t)=s_1(t-\alpha)$
 - If $\alpha < 0$ the signal is shifted to left= advanced
 - If $\alpha > 0$ the signal is shifted to right= delayed
- **Compression/expansion:** serves to represent it over a time interval shorter than that of the original signal (compression), or longer (expansion)
 - $s(t)=s_1(t*\alpha)$
 - If $\alpha > 1$ the signal is compressed
 - If $\alpha < 1$ the signal is expanded

Signal operation



Signal operation

Periodization: The periodization of a signal serves to reproduce the same signal every x seconds. This is useful in signal processing, for example when analyzing signals in the frequency domain using Fourier series.



Time versus frequency

Time Domain:

- Shows amplitude over time
- Useful for waveform inspection

Frequency Domain:

- Graphically represent the frequencies contained in the signal.
- The graphical frequency representation of a signal is called the spectrum of the signal
- The amplitude spectrum is even. It is therefore symmetrical with respect to the ordinate axis.
- The phase spectrum is odd. It is therefore symmetrical with respect to the origin
- Obtained via **Fourier Transform**

Time versus frequency

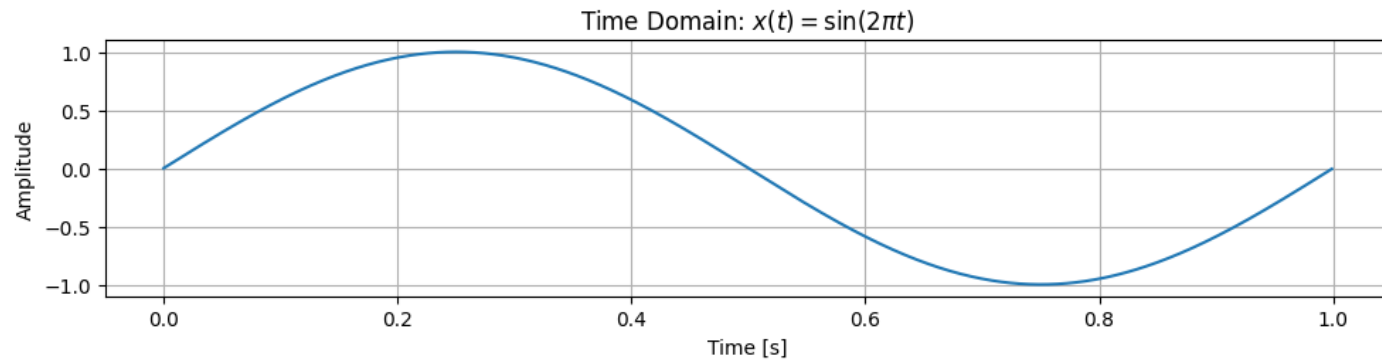
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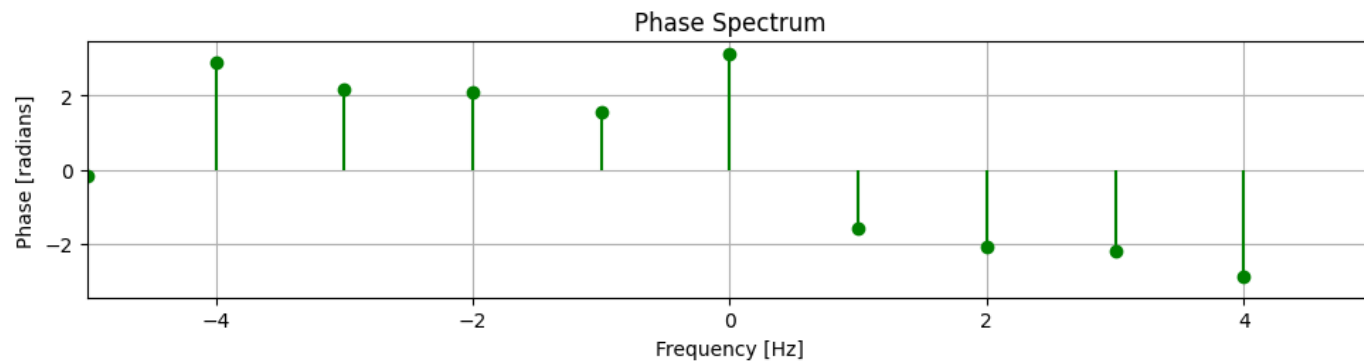
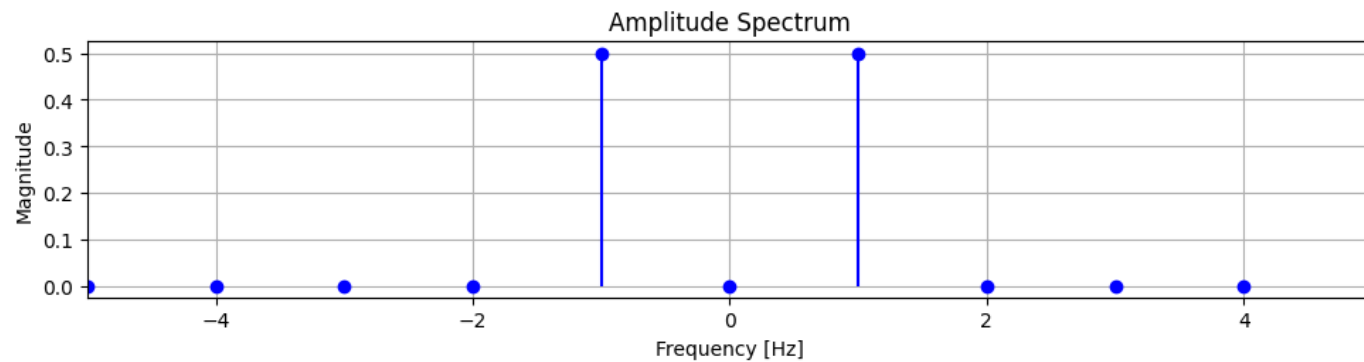
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Time versus frequency



Period 1s $\rightarrow$ Frequency of 1Hz



Why frequency domain?

- **Highlights underlying patterns in signals (e.g., dominant tones in music or voice)**
- **Enables noise removal (e.g., low-pass, high-pass filters)**
- **Crucial for compression algorithms (e.g., MP3, JPEG)**
- **Helps diagnose issues (e.g., detecting faulty machinery from vibrations)**
- **Facilitates feature extraction for AI/ML (e.g., Mel-spectrograms)**

Filters and Smoothing in Signal Processing

A filter is a system or algorithm that selectively allows certain frequencies to pass through while attenuating others. It is a fundamental tool in signal and audio processing

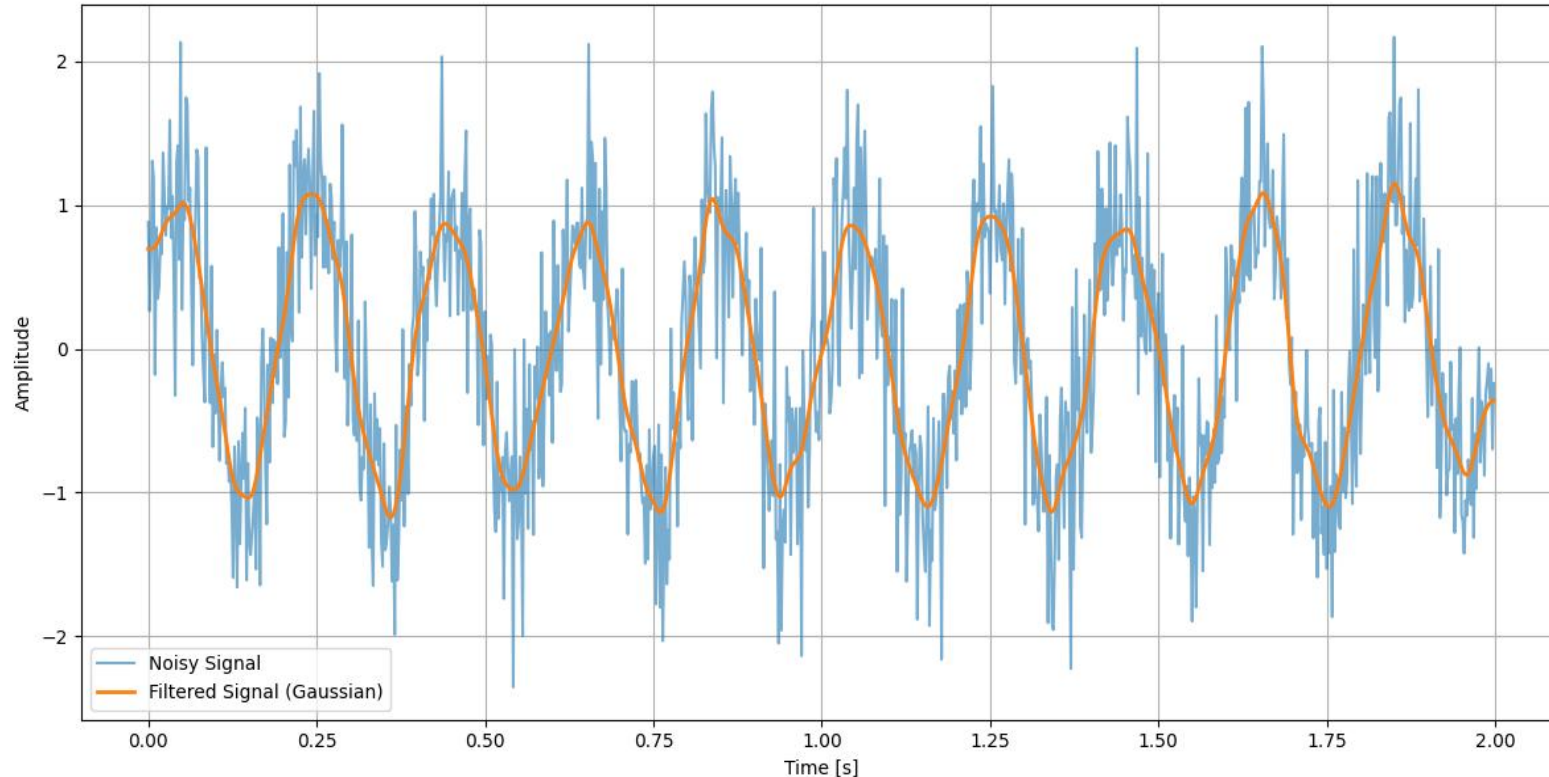
Type	Passes	Blocks	Use Case
Low-pass	Low frequencies	High frequencies	Removing high-frequency noise
High-pass	High frequencies	Low frequencies	Removing drift or DC offset
Band-pass	A specific frequency band	Everything else	Isolating musical notes or channels
Band-stop	Everything except a band	A specific frequency range	Removing hum or specific interference

Example: To remove background hiss from a voice recording, we can apply a **low-pass filter** that cuts off all frequencies above 8 kHz.

Filters and Smoothing in Signal Processing

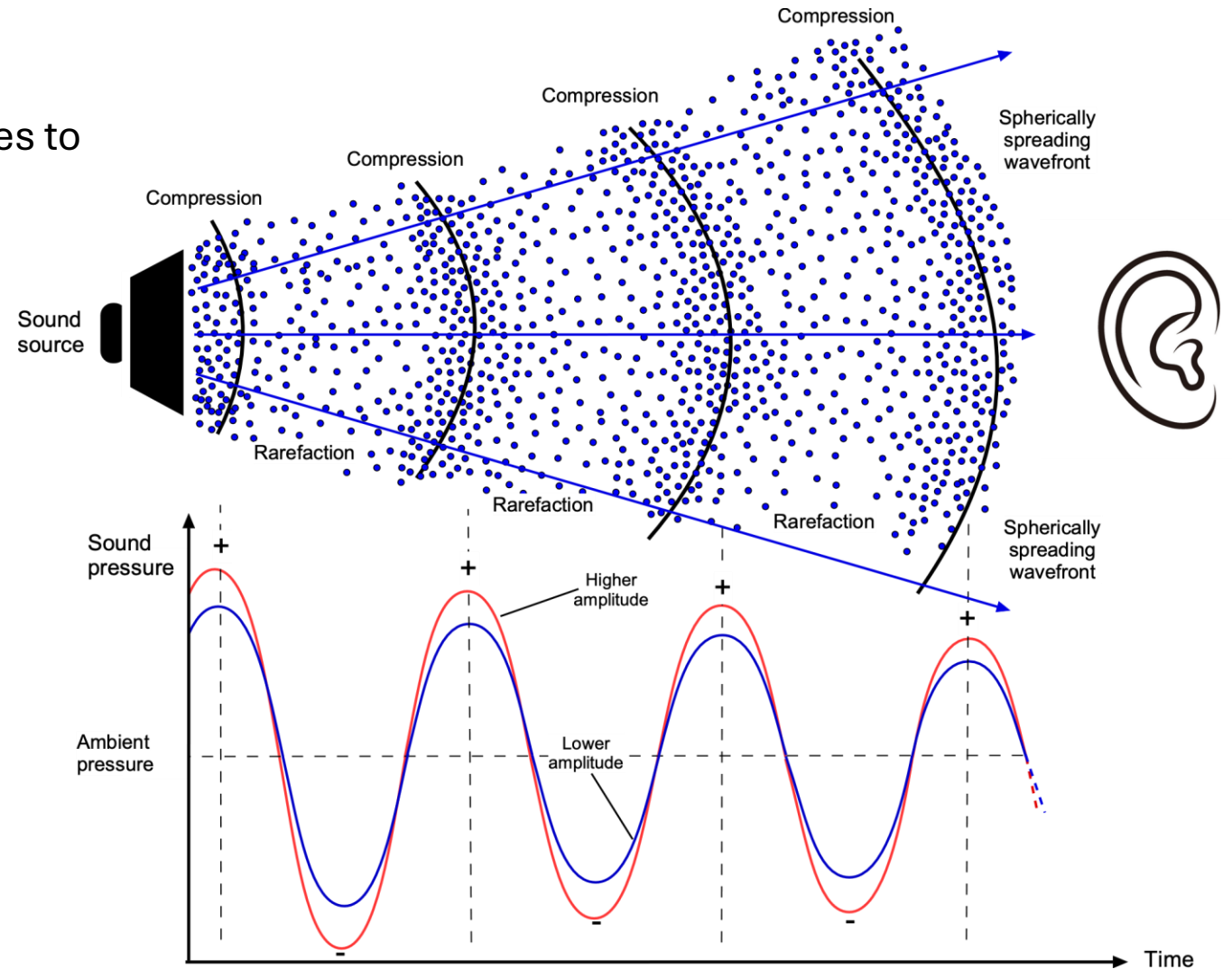
Filters can also operate in the time domain to reduce rapid fluctuations or noise:

Type	Description	Use Case
Moving Average Filter	Averages each point with its neighbors	Smooths short-term fluctuations
Gaussian Filter	Applies a weighted average using a Gaussian kernel	Smooths while preserving transitions

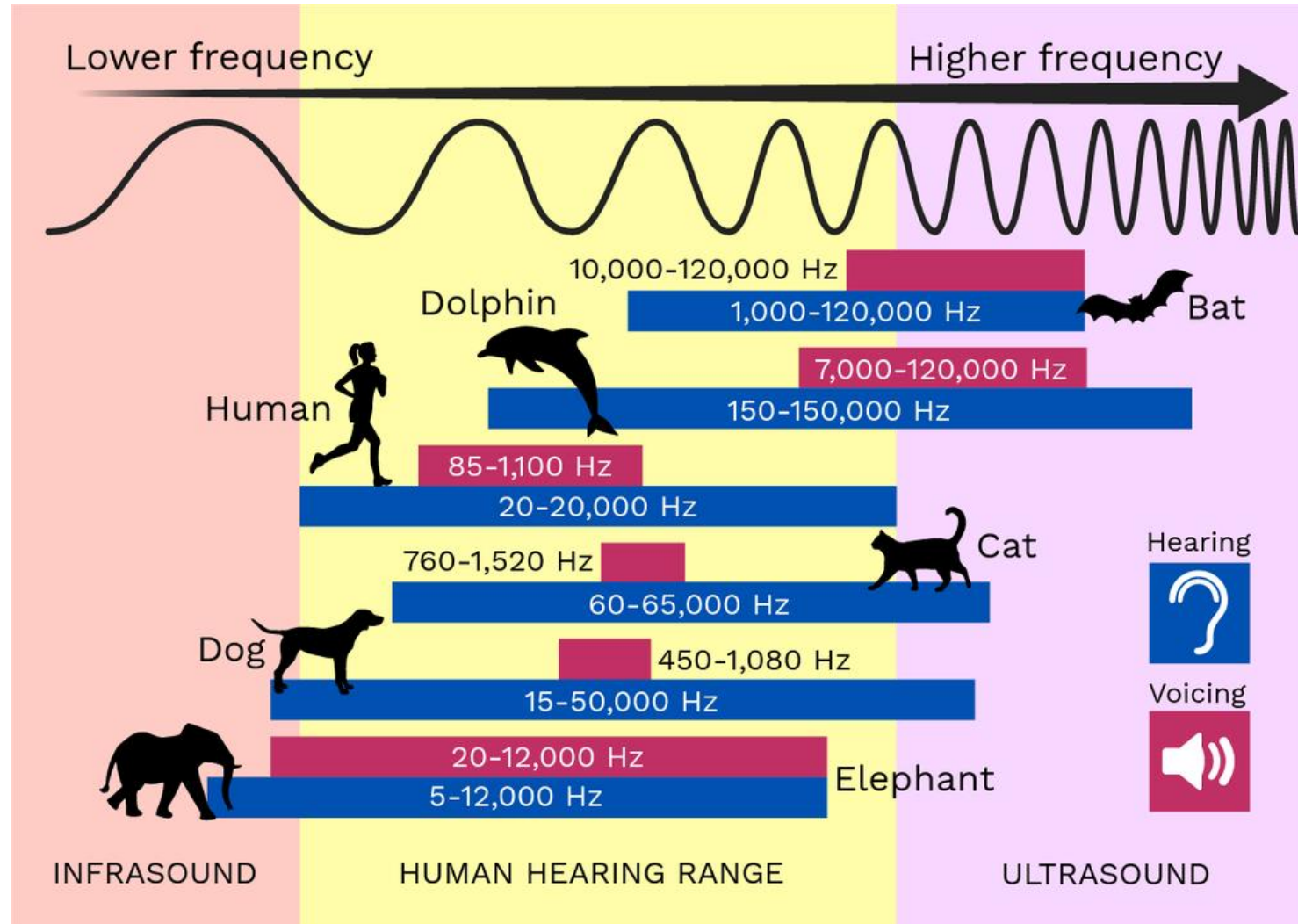


What is the sound?

- Sound is a mechanical wave produced by the vibration of an object which causes air molecules to oscillate.
- It propagates through a medium (like air) as oscillations of pressure.
- These oscillations create compression and rarefaction zones



What is the sound?



What is an audio signal?

Definition:

- An audio signal is a digital or analog representation of sound.
- It contains all the information needed to reproduce the sound.
- Typically stored in the time domain as a waveform.

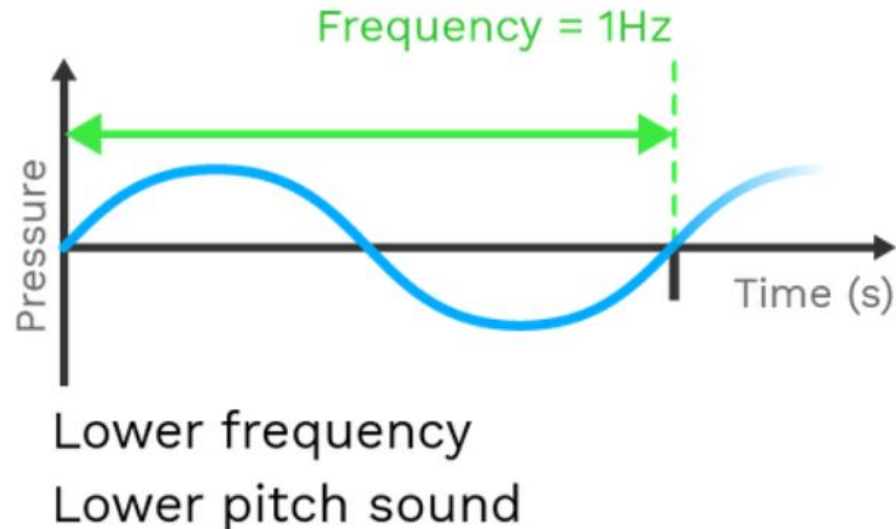
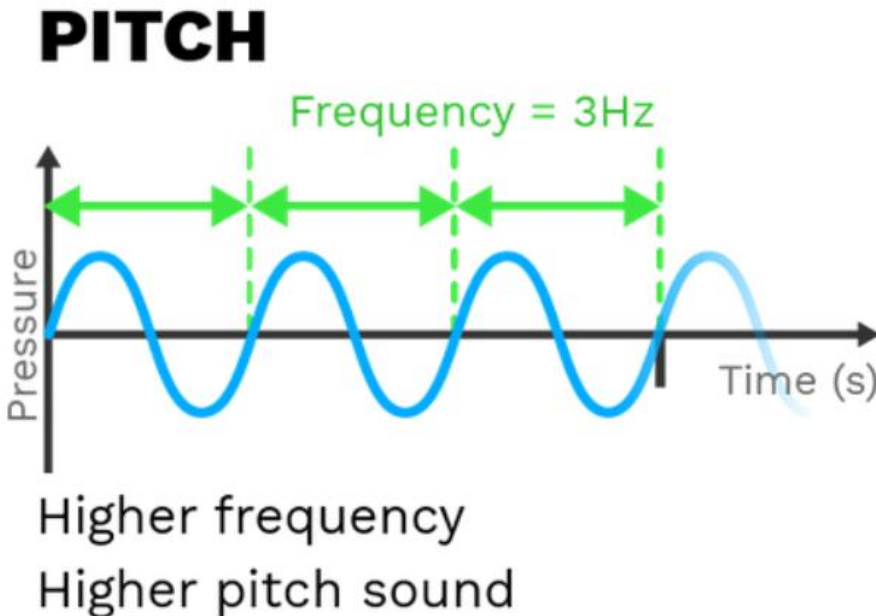
Common properties:

- Frequency
- Amplitude
- Pitch
- Loudness
- Timbre

Pitch

Definition:

- Pitch is the perceptual property of sound that allows us to classify it as "high" or "low".
- Technically, it corresponds to the frequency of the sound wave.
- A higher frequency is perceived as a higher pitch.
- A lower frequency is perceived as a lower pitch.
- Pitch is critical in music and speech processing: Melody, harmony, intonation, and tone depend on pitch



Sound Intensity & Human Perception

Definition:

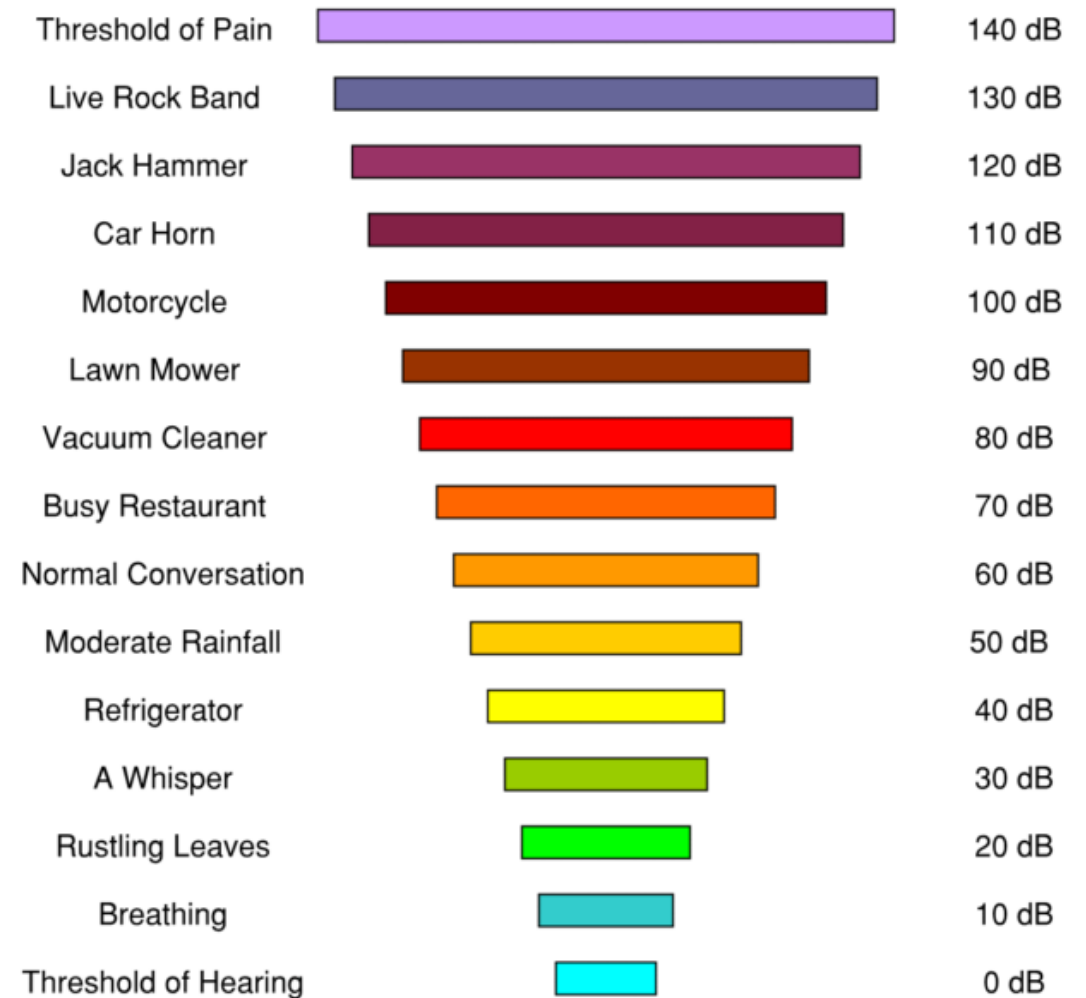
- **Sound Power:** Total sound energy emitted per unit time, measured in watts (W).
- **Sound Intensity:** Sound power per unit area, measured in W/m^2 .
- **Threshold of Hearing (TOH):** The quietest sound a typical human ear can detect. Defined as 10^{-12} W/m^2 .
- **Threshold of Pain (TOP):** The level at which sound becomes painful to hear. Defined as 10 W/m^2 .

Logarithmic Scale & Decibels

- The range of sound intensity humans can perceive spans over 13 orders of magnitude.
- To capture this wide range, we use a logarithmic scale, expressed in decibels (dB).
- A decibel (dB) is a relative unit that expresses the ratio between two values of intensity.
- For sound, we typically compare the intensity to the Threshold of Hearing (TOH).

Formula:

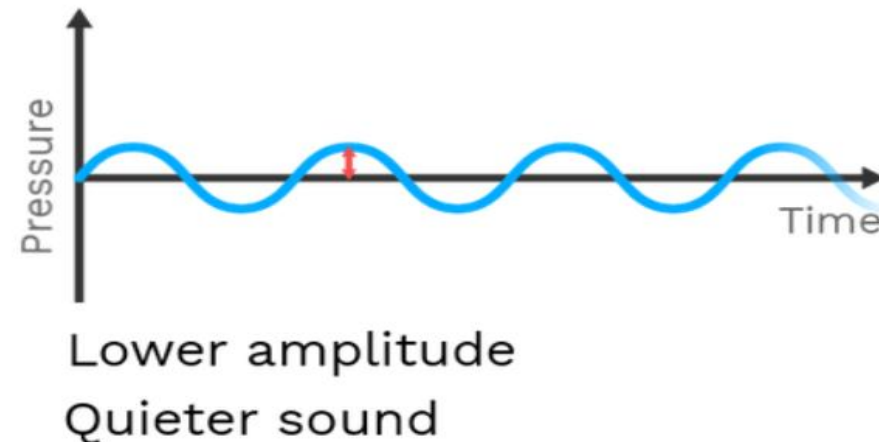
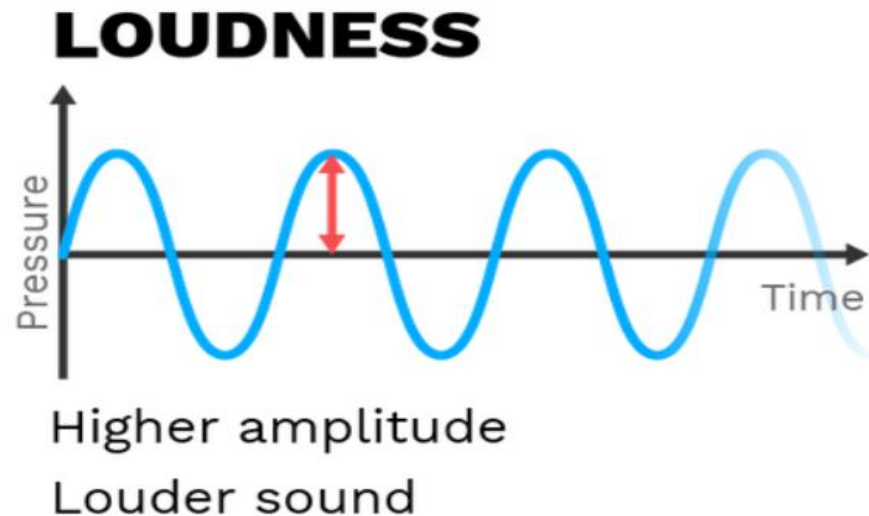
$$\text{dB}(I) = 10 \times \log_{10}(I / I_{\text{toh}})$$



Loudless

Definition:

- Loudness is the **perceived intensity** of sound — how loud or soft a sound feels to the human ear. Technically, it corresponds to the frequency of the sound wave.
- While intensity is a physical property (measured in W/m^2), loudness is a subjective experience.
- It depends not only on the intensity but also on the frequency and the sensitivity of the ear.
- The same intensity can be perceived louder or softer depending on the pitch
- Doubling the perceived loudness $\neq$ doubling the intensity.
- A ~ 10 dB increase is generally perceived as "twice as loud".



Timbre

Definition:

It is the **quality** or **color** of a sound that makes it unique — even when pitch and loudness are the same.

Ex: A **violin** and a **flute** playing the same note at the same volume sound different because of their **timbre**.

Timbre is shaped by:

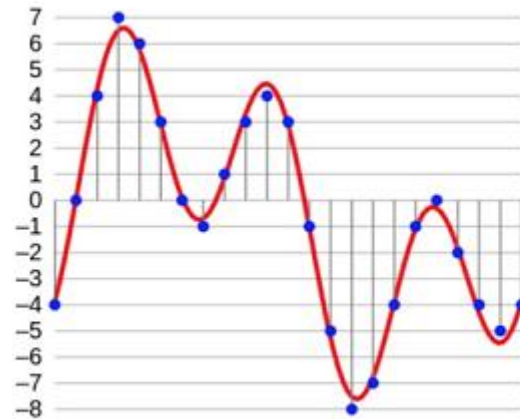
- **Harmonic content:** The number and relative strength of overtones (frequencies above the fundamental). Richer harmonics give a brighter or fuller sound.
- **Envelope (Attack, Decay, Sustain, Release - ADSR):** Describes how the sound evolves over time. A piano and a violin may sustain a note differently.
- **Spectral envelope:** How energy is distributed across the frequency spectrum, affecting the tonal balance and brightness of the sound.

Timbre is critical for:

- Identifying instruments or speakers
- Emotion recognition in speech
- Voice and music classification in ML systems

Digitization

- **Real signals are analog (continuous) in nature**
- **Electronic and computer systems are digital systems**, that is they are able to process information of a digital nature. Need to **convert analog signals to digital signals** in order to be able to process them computationally
- **Digital signals are a sequence of discrete values** and can only take **a finite number of values**.
- This process is called **digitization** or **Analog to Digital Conversion (ADC)** and has main steps:
 - **SAMPLING**
 - **QUANTIZATION**



Sound wave



[-4, 0, 4, 7, 6, 3, 0, -1, 1, 3, 4, 3,
-1, -5, -8, -7, -4, -1, 0, -2, -4, -5, -4]

Array

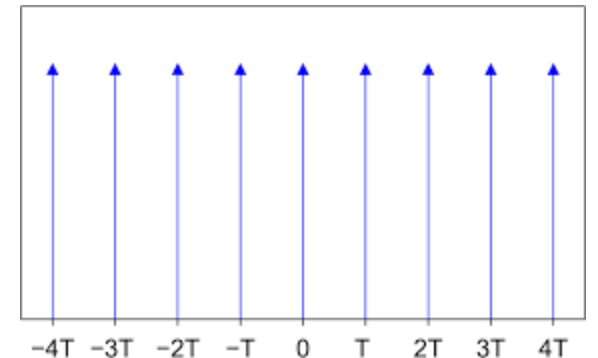
Sampling

Sampling consists of taking at precise times, often regular and equidistant, instantaneous values of the analog signal (audio signal) to be digitized.

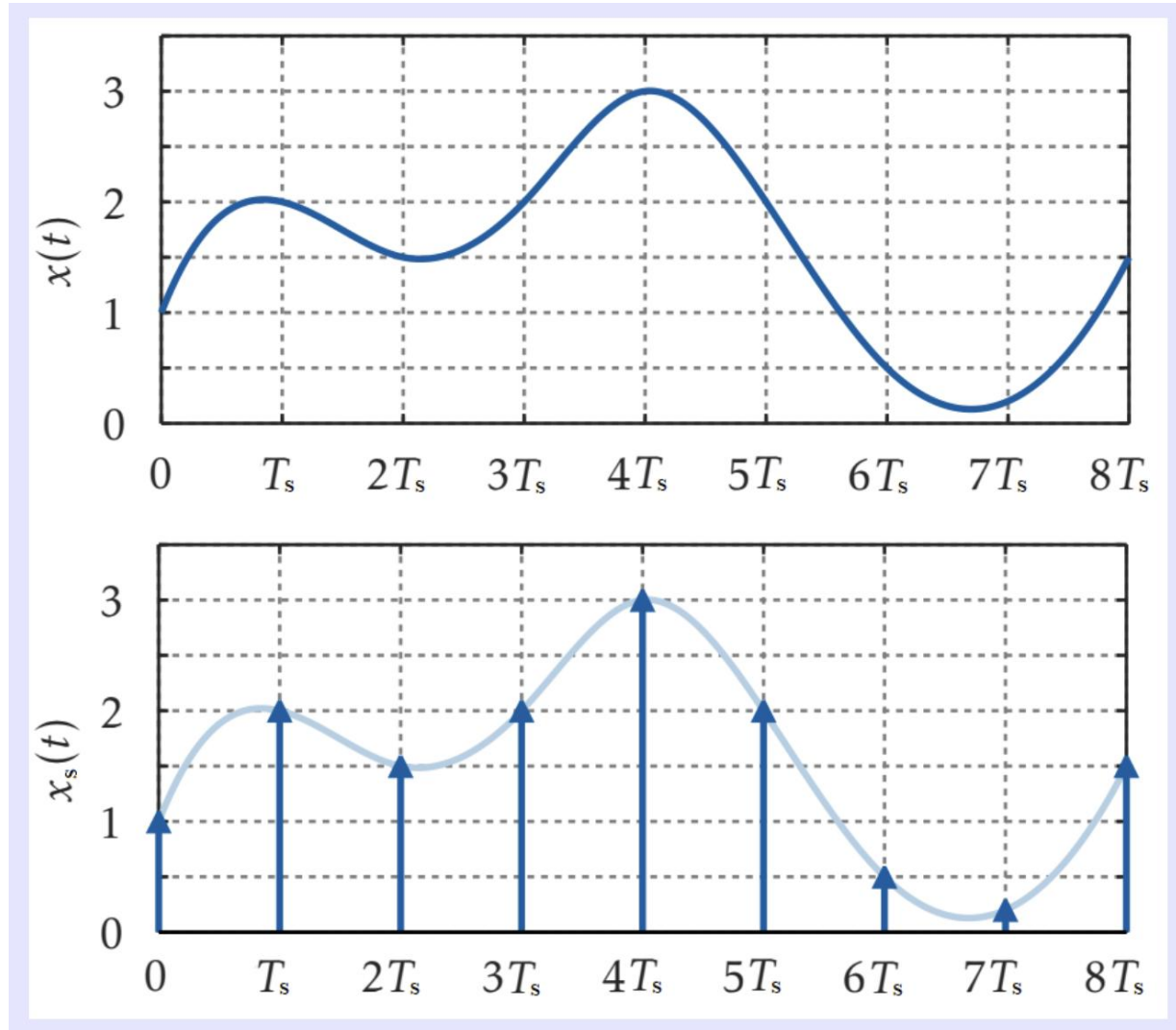
- The time between two samples is called the sampling period, denoted T_s .
- The sampling frequency or sampling rate (F_s) is the number of samples taken per second: $F_s = 1/T_s$
- Smaller T_s (higher F_s) means more accurate signal representation — but also more data
- Sampling inherently introduces sampling error, which depends on the chosen sampling rate.
- Mathematically, ideal sampling is modeled by **multiplying** the continuous signal $x(t)$ by a Dirac comb:

$$x_s(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

Attention: Convolution with a Dirac comb serves to periodize it



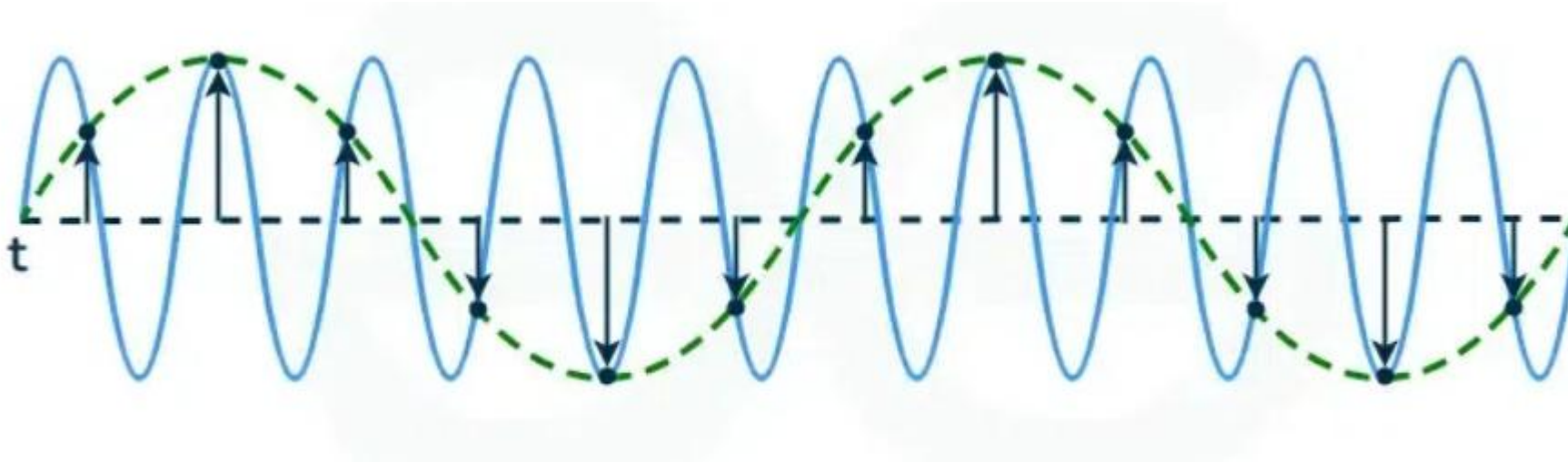
Sampling



Shannon–Nyquist criterion

When we sample a signal, its frequency spectrum becomes periodic (repeats at intervals of the sampling rate).

If the sampling frequency F_s is too low, spectral components overlap — this is called spectral overlap or **aliasing**.



Even though the green dashed curve looks "good", it is not the original signal. It's a false signal introduced by undersampling — this is the alias.

-> **SHANNON-NYQUIST CRITERION**

Shannon–Nyquist criterion

- **Shannon–Nyquist criterion:**

$$F_s \geq 2F_{\max}$$

Where $F_{\max}$ is the highest frequency present in the original signal.

If this condition is not respected, there will be spectral overlap, which will cause the loss of information during the signal restitution.

- **Why does the CD technology use an F_s of 44100Hz?**

Shannon–Nyquist criterion

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- **Why does the CD technology use an F_s of 44100Hz?**

The average human ear can hear frequencies from **20 Hz to 20,000 Hz (20 kHz)**.

So, $2 \times 20\text{kHz} = 40\text{kHz}$therefore, ensures **no aliasing** within the human hearing range.

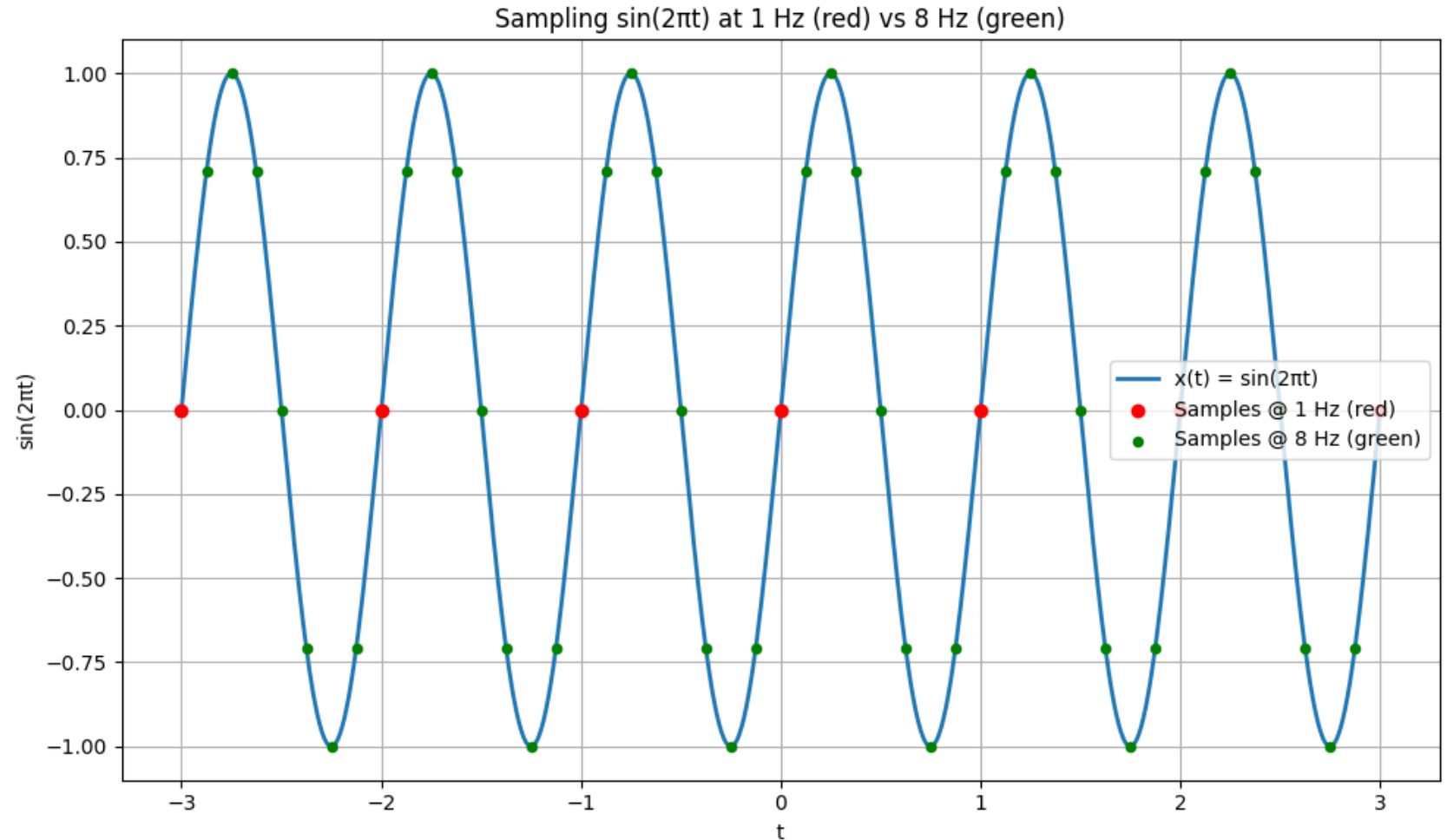
Sampling

Sampling of 1

$F_s = 1\text{Hz} = F_{\text{max}} \rightarrow$ aliasing

Sampling of 8Hz (1pt every 0,125s)

$F_s = 8\text{Hz} \gg F_{\text{max}} \rightarrow$ no aliasing



Quantization

Quantization is the process of mapping each sampled amplitude to the nearest value within a finite set of discrete levels.

- After sampling (taking measurements at regular time intervals), we still have infinite-precision amplitudes.
- Quantization reduces these to a finite number of values → necessary for digital storage.

How It Works

- Choose a **bit depth** (e.g., 16-bit, 24-bit).
- The number of levels is: $\text{Levels} = 2^{\text{bit depth}}$ (e.g., 16-bit = 65,536 levels)
- Each continuous sample is **rounded** to the nearest level.

Bit Depth	Levels	Notes
8-bit	256	Low quality, used in early games
16-bit	65,536	CD quality audio
24-bit	~16M	Professional audio, better dynamics

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Quantization

Sampling times: $t = 0, 0.125, 0.25, 0.375, 0.5, \dots$

Sampled values:

$$x(0) = \sin(0) = 0$$

$$x(0.125) \approx 0.707$$

$$x(0.25) = 1$$

$$x(0.375) \approx 0.707$$

$$x(0.5) = 0$$

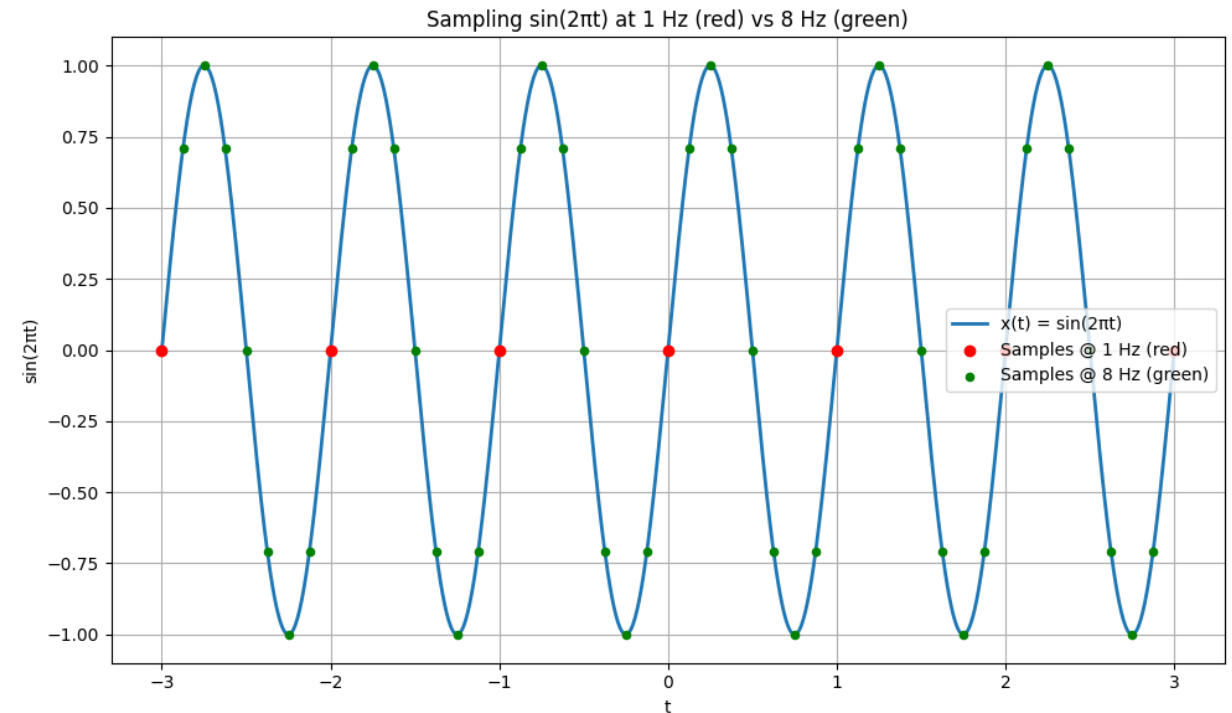
Quantization: Rounding Values to Discrete Levels

Real sampled values: e.g., 0.707, 1.0, 0

Suppose 8-bit quantization $\rightarrow$ 256 levels between -1 and 1

Level spacing ≈ 0.0078

-1.0000, -0.9922, -0.9844, ..., 0.7031, 0.7109, ..., 1.0000



Sampled Value

0.707

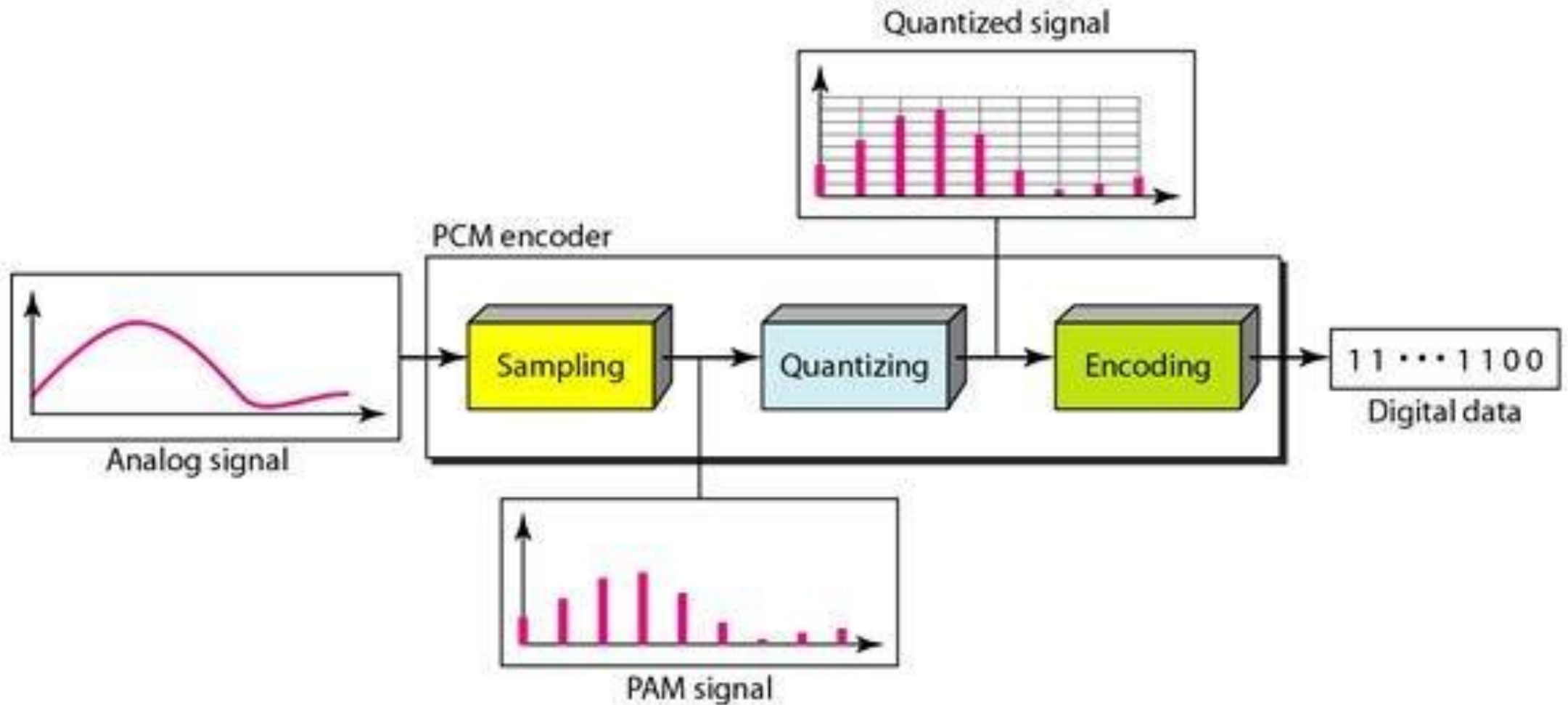
1.0

Nearest Quantized Level

~ 0.7031

1.0

Encoding



Common Audio File Formats

Audio is typically considered **unstructured data**, but there are standard formats for storing and sharing it. Here are the most common:

Format	Description	Compression	Notes
WAV	Waveform Audio File Format (Microsoft/IBM)	✗ None	High quality, uncompressed; large file size.
AIFF	Audio Interchange File Format (Apple)	✗ None	Similar to WAV; also uncompressed.
MP3	MPEG-1 Audio Layer 3 (Fraunhofer Institute)	✓ Lossy	Widely used; smaller files, reduced sound fidelity.
FLAC	Free Lossless Audio Codec (Xiph.Org Foundation)	✓ Lossless	Compresses audio without quality loss.